Multi-curve EUR rate curves and swap pricer

Validation report, generated on 2026-11-04 by `studies/report.py`.

> Levels reconstructed on from public ECB data; QuantLib validation independent of the data source; interpolation and PnL explain studies on synthetic data.

Self-review by the model author, not an independent validation in the SR 11-7 sense: no separation of functions, no review of the usage framework. Every numerical threshold below is recomputed by this script.

1. Recomputed thresholds

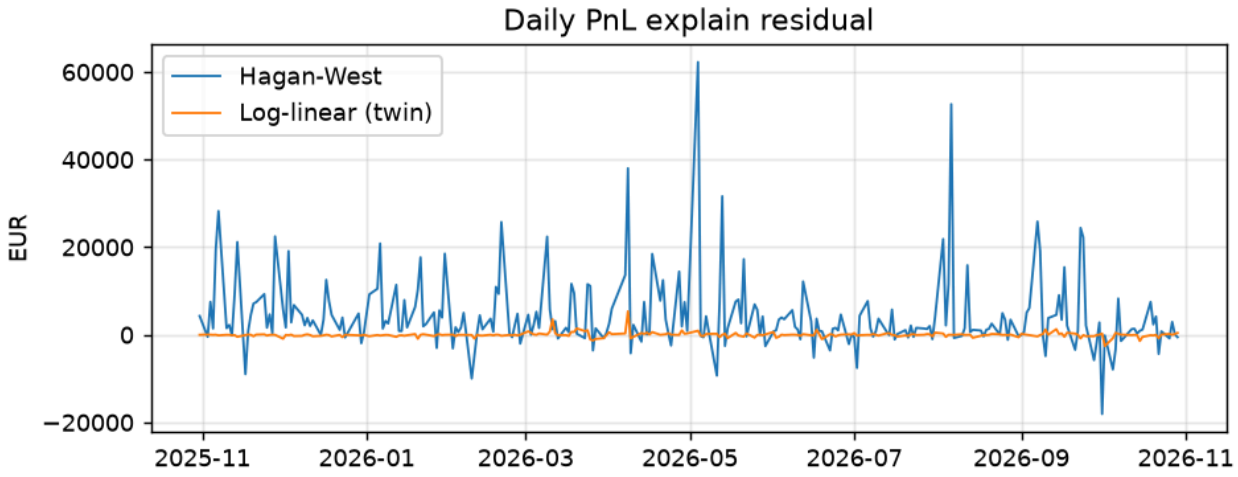
Repricing and QuantLib bench

Bench days: one in ten, month ends and the last day. Maximum gaps on the calibration instruments (repricing, threshold 1e-13) and on the discount factors, every 29 days up to 40 years, against QuantLib (threshold 1e-10). Run time: 44 s.

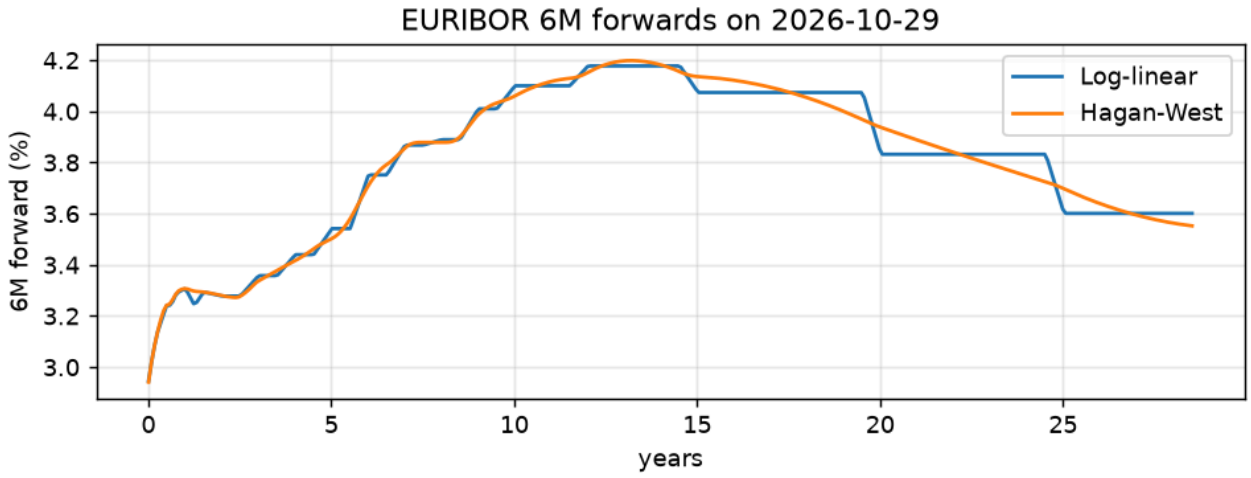
Interpolation	Days	Max repricing	Max €STR DF	Max 6M DF
Log-linear	38	3.4e-14 OK	4.5e-15 OK	6.5e-14 OK
Hagan-West	38	9.5e-14 OK	3.0e-13 OK	1.4e-13 OK

PnL explain

254 days: mean absolute residual = 4.93% of the mean absolute PnL (Hagan-West), 0.26% for the log-linear twin. Morning briefing alert threshold: |residual| > max(10% of PnL, 10 000 EUR); days above: 30.



2. Interpolation



Log-linear on ln P: step-like forwards. Hagan-West (monotone convex): continuous forwards, chosen by the leave-one-out test (section 4). The trade-off: the value is not differentiable in the quotes, hence the PnL explain residual.

3. Morning briefing

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Briefing for 2026-10-29 (previous day 2026-10-28), Hagan-West, reconstructed quotes
Portfolio value: 2 392 338 EUR; DV01 (sum of buckets): 36 256 EUR/bp
Top 3 pillars: IRS 10Y 42 182 EUR/bp; IRS 6Y 21 440 EUR/bp; IRS 3Y -18 227 EUR/bp
Previous-day PnL: total 105 698; carry 283; fixings 0; delta 106 415; gamma -445; residual -555
(EUR)
Alerts: none
Run time: 4.9 s
```

4. Studies

Results produced by `studies/leave_one_out.py`, `nss.py`, `risk.py` and `pnl.py`, included as is.

Leave-one-out: log-linear against Hagan-West

30 days (one quote day in ten + the 5 days of the 10Y IRS basis jump), noisy quotes. Gap |implied rate − quote| of the removed instrument, in bp.

Curve	Interpolation	Mean	Max	Jump: mean	Jump: max	Mean roughness (bp²/year)
€STR	log-linear	1.36	5.46	1.59	4.80	4.7
€STR	Hagan-West	0.32	1.37	0.40	1.37	0.7
6M	log-linear	2.02	12.69	2.41	8.96	8.9
6M	Hagan-West	0.62	3.70	0.96	3.12	1.1

Per removed instrument (all days, mean | max):

Curve	Instrument	LL mean	LL max	HW mean	HW max
€STR	2W	0.95	2.25	0.16	0.43

€STR	1M	2.23	4.80	0.25	0.65
€STR	2M	1.63	3.49	0.19	0.95
€STR	3M	2.47	5.08	0.22	0.88
€STR	6M	2.40	5.46	0.25	0.62
€STR	9M	0.83	1.82	0.24	0.87
€STR	1Y	0.70	1.47	0.26	1.30
€STR	18M	0.81	1.95	0.29	0.94
€STR	2Y	1.11	3.00	0.29	0.97
€STR	3Y	2.19	4.35	0.40	1.37
€STR	4Y	2.20	3.27	0.31	0.69
€STR	5Y	1.71	2.36	0.27	0.70
€STR	6Y	1.41	2.10	0.29	0.66
€STR	7Y	1.00	1.44	0.23	0.52
€STR	8Y	0.74	1.23	0.22	0.80
€STR	9Y	0.52	1.35	0.22	0.94
€STR	10Y	0.66	1.16	0.26	0.95
€STR	12Y	0.87	1.34	0.58	1.28
€STR	15Y	0.42	0.88	0.50	1.02
€STR	20Y	1.73	2.38	0.41	1.23
€STR	25Y	1.94	2.71	0.83	1.35
6M	1x7	0.59	1.21	0.48	1.02
6M	2x8	0.64	1.32	0.42	0.80
6M	3x9	1.09	2.31	0.62	1.26
6M	6x12	5.30	12.25	1.02	2.68
6M	9x15	4.69	10.85	1.06	2.90
6M	12x18	5.38	10.72	1.09	3.70
6M	18x24	4.41	12.69	0.94	2.51
6M	3Y	2.11	4.26	0.37	1.00
6M	4Y	2.21	3.33	0.40	0.95
6M	5Y	1.85	2.87	0.38	1.27
6M	6Y	1.34	2.14	0.38	1.00
6M	7Y	1.03	1.74	0.38	0.74
6M	8Y	0.81	1.80	0.40	1.38
6M	9Y	0.75	2.59	0.59	2.66
6M	10Y	0.99	2.36	0.72	3.12
6M	12Y	1.03	2.36	0.79	2.38
6M	15Y	0.39	1.18	0.51	1.46
6M	20Y	1.82	2.62	0.56	1.50
6M	25Y	1.90	2.76	0.76	1.60

NSS fitted to the bootstrapped zeros (Hagan-West)

255 days of noisy quotes; noiseless control one day in ten (26 days). Fit to the pillar zeros, equal weights.

Curve	Quotes	Mean RMSE (bp)	Max RMSE (bp)	Max residual (bp)
€STR	noisy	0.55	1.86	6.17
6M	noisy	0.67	1.87	6.02
€STR	noiseless	0.44	1.61	5.46
6M	noiseless	0.50	1.70	5.34

Day-to-day stability (noisy quotes): standard deviation of daily changes, in bp, and number of days where a λ moves by more than 50%.

Curve	Δ 10Y zero	Δ β0 (level)	Δ β1 (−slope)	Δ β2 (curvature)	Δ β3	λ jumps
€STR	3.4	52.8	52.9	50.6	211.6	19
6M	3.5	75.8	76.6	82.0	288.9	32

Extrapolation at 40 years, NSS minus extended flat forward (bp): €STR mean -5.5, min -16.6, max +15.9; 6M mean -5.7, min -17.3, max +15.9

Risk of the swap portfolio on 2026-10-29

Ten swaps (payers and receivers, forward starts, seasoned swaps with a past fixing), `data/portfolio.csv`. Past EURIBOR 6M fixings: reconstructed DEPOSIT 6M quote of the fixing day (assumption, see `assumptions.md`).

Measure	Hagan-West	Log-linear
Value (EUR)	2,392,338	2,403,363
Quote DV01, ±1 bp rebootstrap (EUR/bp)	36,256	36,202
Σ buckets (EUR/bp)	36,256	36,202
Zero DV01, +1 bp on zero rates (EUR/bp)	37,593	37,590
Bucket run time (s)	0.59	0.20
Max hedge residual (EUR/bp)	1.8e-12	4.5e-13

Quote DV01 and zero DV01 differ (~4%): one bp on the quotes does not shift the zeros by one bp (annuity against zero duration, and the €STR/6M basis moves with it).

Buckets and hedge (Hagan-West)

Hedge in calibration instruments at par, fixed-rate payer notional (negative = receiver). Instruments with |bucket| > 100 EUR/bp:

Instrument	Bucket (EUR/bp)	Hedge notional (EUR)
IRS 10Y	42,182	-50,133,703
IRS 6Y	21,440	-39,776,038
IRS 3Y	-18,227	64,539,777
IRS 30Y	18,153	-10,000,022

IRS 8Y	15,597	-22,416,108
IRS 5Y	-12,476	27,325,610
IRS 12Y	-10,733	10,999,803
IRS 4Y	-9,619	25,907,706
IRS 15Y	6,502	-5,610,761
IRS 7Y	5,974	-9,653,432
IRS 9Y	-5,272	6,847,970
FRA 6x12	-5,265	106,098,819
FRA 12x18	-3,235	66,941,728
DEPOSIT 6M	-2,845	57,071,904
FRA 18x24	-2,704	56,212,683
FRA 1x7	-1,619	32,557,361
OIS 12Y	-483	487,737
IRS 20Y	364	-256,335
OIS 7Y	-213	338,510
OIS 4Y	-175	465,238
OIS 10Y	-166	194,464
OIS 8Y	-165	233,155
OIS 6Y	-156	286,136
OIS 9Y	-136	173,879
OIS 9M	-121	1,630,008
OIS 5Y	-114	246,828

IRRBB, ΔEVE (BCBS 2016, EUR, EBA floor)

Scenario	Hagan-West	Log-linear
parallel_up	5,942,941	5,942,396
parallel_down	-9,557,624	-9,556,810
short_up	-3,529,459	-3,514,878
short_down	3,665,135	3,642,874
steepener	6,757,166	6,735,581
flattener	-6,076,958	-6,060,827

Worst scenario: parallel_down (-9,557,624 EUR). Shocks exported for project 04 in `studies/irrbb_shocks.yaml` (bp per maturity, before the floor).

Bucket locality: swap P10Y-2024 alone (payer, EUR 25m, 2.80%, maturity 2034-03-15)

A spot swap of quoted tenor (P10Y) has its dates on the pillars: its buckets are identical under both interpolations (IRS 10Y + OIS). P10Y-2024 has its dates between pillars, so the interpolation matters.

Instrument	Hagan-West	Log-linear
OIS 1W	-1.5	-0.8
OIS 2M	-1.9	-0.1

OIS 3M	7.6	4.8
OIS 6M	26.2	25.5
OIS 9M	-13.5	-15.8
OIS 1Y	-60.4	-51.4
OIS 18M	75.8	73.6
OIS 2Y	-82.9	-75.1
OIS 3Y	-16.9	-41.4
OIS 4Y	-75.4	-57.3
OIS 5Y	-61.4	-69.6
OIS 6Y	-79.5	-86.9
OIS 7Y	-119.0	-157.2
OIS 8Y	-34.9	20.0
OIS 9Y	6.4	-0.0
OIS 10Y	-2.0	-0.0
DEPOSIT 6M	-1,681.6	-923.8
FRA 1x7	759.9	-0.0
FRA 2x8	-6.3	-0.0
FRA 3x9	21.1	19.6
FRA 6x12	-11.2	-15.2
FRA 9x15	4.5	10.2
FRA 12x18	-1.5	-5.3
FRA 18x24	-4.1	-0.2
IRS 3Y	4.4	1.8
IRS 4Y	2.0	-1.1
IRS 6Y	-121.8	2.3
IRS 7Y	5,981.4	9,753.1
IRS 8Y	15,578.0	6,427.4
IRS 9Y	-5,257.3	-0.0

Share of |buckets| outside IRS 7Y and 8Y: Hagan-West 28.4%, log-linear 9.3%; instruments above 1 EUR/bp: 30 against 22.

Daily PnL explain, 2025-10-31 → 2026-10-29 (254 days)

Synthetic market: quotes reconstructed from the ECB and noised (0.2 to 0.5 bp), EURIBOR 6M fixings reconstructed (DEPOSIT 6M quote of the day). The fixings component is built by assumption. This result validates the decomposition mechanics, not market behaviour.

Component	Mean (EUR)	Mean absolute value (EUR)	Max absolute (EUR)
total	1,404	110,972	478,072
carry	367	385	1,510
fixings	0	2	163
delta	-2,819	110,920	473,019
gamma	-465	465	9,668

residual	4,319	5,468	62,247

- Mean absolute residual: 4.93% of the mean absolute PnL; day-by-day median of |residual| / |PnL|: 3.15%; explained variance 1 − var(residual) / var(PnL): 99.6345%.
- Worst day: 2026-05-04 (residual 62,247 EUR, PnL 244,111 EUR).
- **Source of the residual**: the non-differentiability of Hagan-West. Its monotone convex scheme changes region when quotes move, V is not smooth and the first-order expansion breaks down (on 2026-08-06, the slope along the path q_c → q1 goes from −54,610 to +9,128 EUR for a move of about 1 bp). Log-linear twin, same days: mean |residual| 289 EUR, max 5,366 EUR (0.26% of the mean absolute PnL).
- Fixing days (non-zero component): 7.
- 10Y IRS basis jump (+2.5 bp from 2026-05-04, 5 days): max |residual| 62,247 EUR on those days. An isolated move of one quote pushes the Hagan-West curve out of its region, hence the worst day of the year.
- Run time: 977 s for 254 days (3.8 s per day, Hagan-West, five curve builds per day).

Full series: `pnl_series.csv`.

Limitations

- Reconstructed quotes (ECB AAA curves, €STR, monthly bases), not real market quotes.
- Past EURIBOR 6M fixings reconstructed from the DEPOSIT 6M quote of the day.
- No convexity adjustment, no CSA other than EUR cash, no 3M curve.